

Poultry, Eggs

In prospective observational studies, poultry intake appears generally neutral for CVD and DM risk (see Fig. 29.2).^{77,78} When combined with its relatively low content of bioactive nutrients, these findings suggest that poultry consumption has minimal cardiometabolic effects. Eggs appear similarly neutral for CVD risk in general populations (see Fig. 29.2).^{79–81} Overall evidence suggests minimal cardiometabolic effects of poultry intake or occasional egg intake (e.g., up to 2 to 3 per week); consistent with recent similar conclusions on dietary cholesterol¹¹ (see Dietary Cholesterol, below).

Dairy

Cardiometabolic effects of dairy foods have generally been considered in relation to a limited set of nutrients: saturated fat, calcium, vitamin D. Yet, more diverse dairy nutrients and processing methods appear relevant. These include fermentation of yogurt and cheese (creating menaquinones), branch-chain, odd-chain, and medium-chain fatty acid contents, probiotics in yogurt, milk fat globule membrane content, and more.^{82,83} In long-term cohorts, milk intake associates with lower risk of stroke, cheese with lower risk of CHD, and yogurt, cheese, and possibly butter with lower risk of DM (see Fig. 29.2).^{84–86} In randomized trials, milk or total dairy consumption increases lean mass and reduces body fat and waist circumference.⁸⁷ In long-term observational studies, yogurt associates with relative weight loss,¹⁵ potentially related to probiotic-microbiome interactions.^{88–91}

In contrast, lower dairy fat content (reduced-fat or non-fat) does not consistently relate to lower cardiometabolic risk. Indeed, in a pooled analysis of 16 global cohorts, individuals with higher blood biomarkers of dairy fat consumption had significantly lower incidence of DM.⁹² Recent experimental evidence suggests that odd-chain saturated fats such as 15:0, fairly unique to dairy fat, reduce inflammation and dyslipidemia by binding to key metabolic regulators and repairing mitochondrial function.⁹³ In sum, current evidence supports recommendations for modest dairy intake (2 to 3 servings/day), especially of yogurt and cheese, with more research needed to define the relevant active ingredients and health effects of dairy fat.

Plant Oils

While “industrial” or “refined” plant oils, especially those rich in n-6 PUFA, have been theorized to be harmful to health, little evidence supports this concern. The great majority of interventional and observational studies of total PUFA, n-6 PUFA, and MUFA have assessed industrialized plant oils like soybean, canola, and safflower oil. The cumulative evidence supports clear benefits of such plant oils (see Macronutrients, below). Whether virgin versions of these oils would have even greater benefits, for example due to greater concentrations of phenolics and phytosterols,^{94,95} is possible and requires investigation. Few studies have evaluated tropical oils (e.g., palm, coconut), beyond their known effects on blood lipids (raising both LDL-cholesterol and HDL-cholesterol).^{96–98}

BEVERAGES

Sugar-Sweetened Beverages

Ecologic data, prospective cohorts, and trials provide convincing evidence that SSB intake increases adiposity. In the United States, calories consumed from beverages rapidly increased after the 1960s, doubling to 21.0% of all calories consumed by 2002—an increase of 222 daily kcal per person.⁹⁹ This was largely due to increased SSBs (sodas, energy drinks, sweetened ice teas, fruit drinks), although that peak has been followed by gradual declines in SSB consumption thereafter.^{100,101} Per serving, SSBs more strongly associate with long-term weight gain than nearly any other dietary factor.¹⁵ Randomized trials confirm that reducing SSBs decreases weight gain and fat accumulation.^{102,103} Calories in liquid form, compared with solid foods, appear to be less satiating and increase total calories consumed.¹⁰⁴ SSB intake also associates with significantly higher incidence of CVD and especially DM (see Fig.

29.3,^{26,105} likely related to weight gain and other additional harms of the high sugar and glycemic load. Given clear evidence for harms, and multiple alternatives (e.g., water, seltzer, unsweetened tea, diet soda, milk), SSBs should be largely eliminated from the diet.

Alternative sweeteners can be artificial (e.g., saccharin, aspartame) or naturally low-calorie (e.g., stevia). Based on observational studies and clinical trials, alternative sweeteners are better for cardiometabolic health than added sugars.^{15,106} Yet, alternative sweeteners may not be completely benign: animal experiments and limited human data suggest influences on brain reward, taste perception, oral-gastrointestinal taste receptors, glucose-insulin and energy homeostasis, metabolic hormones, and the gut microbiome.^{107–111} For instance, if a child becomes accustomed to intense sweet tastes, will that reduce attractiveness of naturally sweet foods such as apples or carrots? In sum, alternative sweeteners can be a useful bridge to eliminate SSBs, but should not be considered innocuous; and subsequent shifts to non-sweetened drinks (e.g., seltzer, tea) should be encouraged.

Milk

See Dairy.

Coffee, Tea

While coffee and tea provide caffeine, these plant extracts—derived from beans and leaves—contain other bioactive compounds. Unrelated to caffeine content, frequent coffee intake (e.g., 3 to 4 cups/day) associates with less insulin resistance, DM, CVD, and in a few studies, heart failure (see Fig. 29.3).^{112,113} Yet, clear physiologic benefits have not been established to support these observations. Acutely, caffeinated coffee worsens BP, insulin resistance, and glucose intolerance.^{114,115} Longer-term, habitual coffee intake does not affect BP, endothelial function, nor glucose metabolism, suggesting tachyphylaxis and/or other partly offsetting factors.^{116,117} In mendelian randomization analyses, genetic variants linked to coffee intake do not associate with CHD, stroke, or DM.^{118–120} Based on lack of physiologic benefits in trials or lower risk in mendelian randomization studies, the cardiometabolic benefits of coffee intake should still be considered questionable.

Like coffee, frequent tea drinking (e.g., 3+ cups/day) associates with lower CVD and DM, with borderline statistical significance (see Fig. 29.3).^{121,122} In randomized trials, certain types of tea modestly lower BP (green, black, sour), LDL-cholesterol (green), and fasting glucose (green, sour).^{116,123–126} In contrast to coffee, these physiologic effects support potential causal cardiometabolic benefits of tea, but larger and more rigorous studies are still needed.

Alcohol

See Chapter 84.

MACRONUTRIENTS

Carbohydrates

For decades, carbohydrates were recommended as the foundation of a healthful diet, e.g., grain products at the base of the 1992 Food Guide Pyramid. Current science makes clear that the quality and food source, rather than total amount, of carbohydrate is most relevant for cardiometabolic health (Fig. 29.4).⁴ (also see Whole Grains, Refined Grains, Starches, Sweets). Because most carbohydrate in modern diets derives from refined starches and sugars, a “low-carb” diet will often produce metabolic benefit. Yet, healthful carbohydrate-containing foods like fruits, legumes, and minimally processed whole grains should not be avoided. For most patients with metabolic risk factors, a focus on reducing refined grains, starches, and added sugars should be a priority.^{4,8,11}

While marketing claims are often made about different forms of sugars (disaccharides), major types (cane sugar, beet sugar, high-fructose corn syrup) are molecularly similar: about half glucose and half fructose. Thus, health differences between total free sugars, added sugars, or different subtypes of sugars have not been clearly demonstrated.^{127–129}